Bounding the Space of Effective Field Theories

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Outline

Introduction

Complex analysis

Bootstrap

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Introduction/Goals

Questions:

1. Is it possible to constrain the *physical* scattering amplitudes $\mathcal{M}(s,t,u)$ using general arguments?
   - Poincaré invariance (translations + rotations + boosts)
   - Crossing symmetry
   - Positivity: $0 \leq \mathrm{Im}\,[\mathcal{M}]$
2. Is there a framework to do this numerically?
3. Can the numerical results be derived from analytical amplitudes?

Answers: Yes, yes, yes

Introduction/Context

We consider the scattering amplitude $\mathcal{M}(s,t,u)$ for **massless**, **identical**, **scalar** particles.

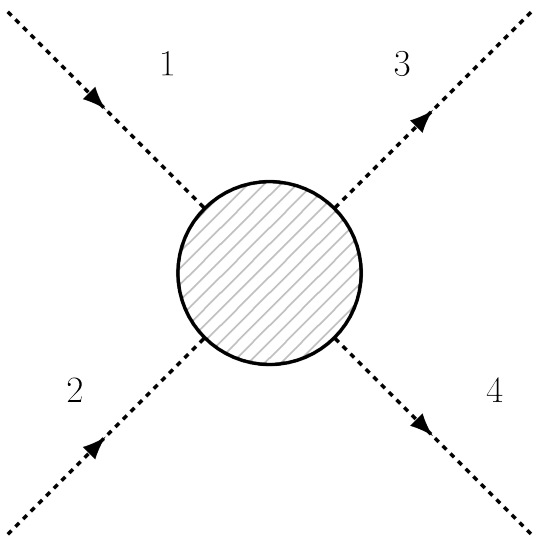

Figure 1: $1+2 \to 3+4$ process.

- Massive particles of mass m with spin J may be produced if the energy is sufficiently large
- Details of the intermediate processes are largely unknown

Introduction/IR and UV bootstrap

There are two interesting limits:

1. Low-energy scattering (centre of momentum energy $\ll M$)
2. High-energy scattering (centre of momentum energy $\gg M$)

where M is some cut-off energy.

1. Low-energy (IR) $\implies$ effective field theory (EFT):

$$\mathcal{M} \approx \mathcal{M}_{\text{low}} = \sum_{\text{powers}} \text{Taylor polynomial of } (s,t,u)$$

2. High-energy (UV) $\implies$ partial wave expansion:

$$\mathcal{M} = \sum_{\text{spins}} \text{Gegenbauer polynomials of } (s,t,u)$$

We show there is a way to relate these two expansions by treating $\mathcal{M}(s,t,u)$ as an analytic function.

Introduction/Not all EFTs are allowed

- **Question**: Are all EFT expansions physically realizable?
- **Answer**: No! They are constrained by the *analytical structure*: $0 \leq \text{Im}\,[\mathcal{M}]$.
- **So what?** The technology we develop allows us to rule-out candidate scattering amplitudes.

Example: EFT with a mass gap M in 3+1 space-time

$$\mathcal{M}(s,t,u) = \underbrace{\frac{1}{M^4}}_{g_2}(s^2+t^2+u^2) + \underbrace{\frac{-10}{M^6}}_{g_3}(stu) + \underbrace{\frac{1}{3M^8}}_{g_4}(s^2+t^2+u^2)^2$$

We will show that these g_n's are **NOT** physically permitted.

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Complex analysis/Cauchy's integral theorem

Suppose that $\mathcal{M}(s,u)$ is an analytic function in s with

- Poles at $s=0$
- Branch cut on the real s axis above the threshold: $s > M^2$

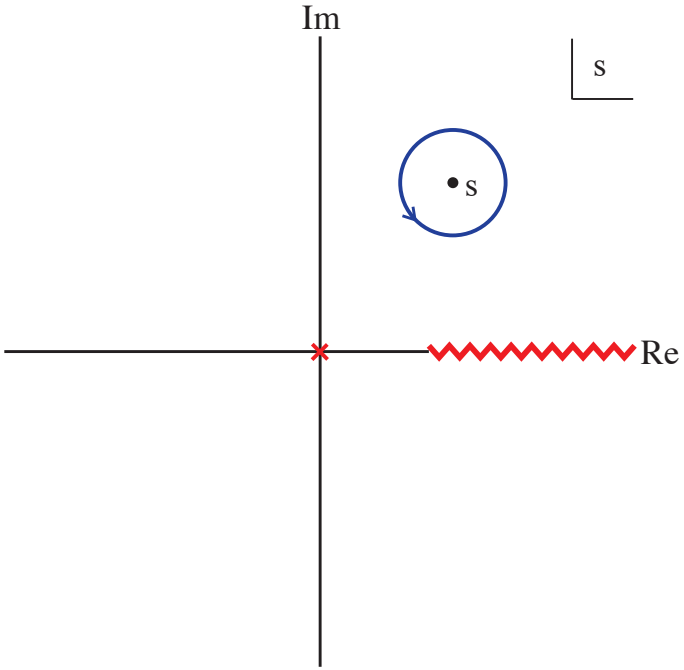
Then, we can use Cauchy's integral theorem:

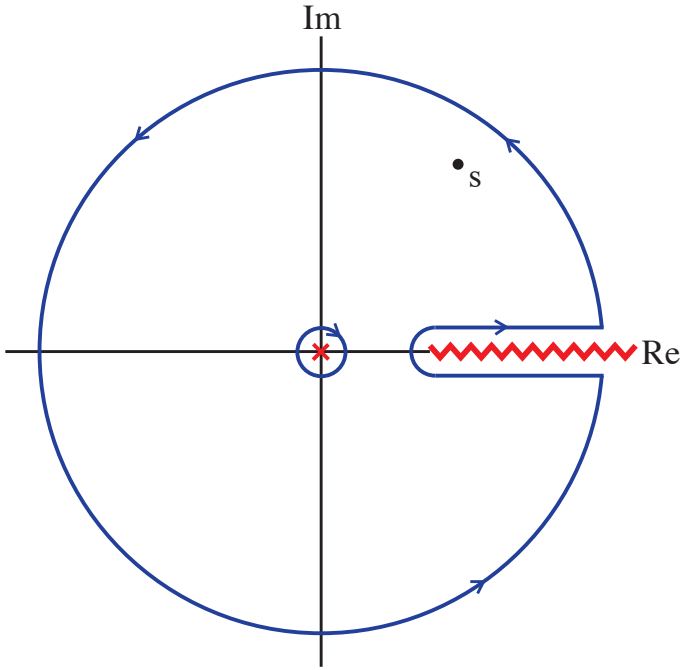
$$0 = \frac{1}{2\pi i} \oint \mathcal{M}(s,u) ds.$$

where the integral is taken in around some closed analytic region.

Complex analysis/Cauchy's integral theorem cont'd

$$0 = \frac{1}{2\pi i} \oint \mathcal{M}(s,u) ds.$$





$$0 = -\text{Res}\left[\mathcal{M}(s,u)\right]_{s=0} + \frac{1}{2\pi}\int_{M^2}^{\infty} ds \text{Disc}_s\left[\mathcal{M}(s,u)\right]$$

- Residue $\implies$ EFT expansion
- Branch-cut $\implies$ partial wave-expansion

Complex analysis/Dispersion relations

EFT: crossing symmetry + massless particles $\Longrightarrow$

$$\mathcal{M}_{\text{low}}(s,u) = -\,g^2\left[\frac{1}{s}+\frac{1}{t}+\frac{1}{u}\right]-\lambda$$
$$+\,g_2(s^2+t^2+u^2)+g_3(stu)+g_4(s^2+t^2+u^2)^2+\ldots$$

Partial-wave:

$$\mathcal{M}(s,u) = s^{\frac{4-d}{2}}\sum_{J\ \text{even}}\mathcal{N}_J c_{s,J}\tilde{C}_J^{\left(\frac{d-4}{2}\right)}\left(1+\tfrac{2u}{s}\right),$$

where $c_{s,J}$ characterize the scattering process. All we can infer is that $0 \leq \text{Im}\left[c_{s,J}\right] \leq 1$

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Bootstrap/B_k sum rules

Plugging in the EFT and Gegenbauers allows to calculate the $B_2, B_4, B_6, \cdots$ that connect the g_n and Gegenbauers

$$\text{Res}\left[\left(\frac{2s+u}{s(s+u)}\right)\left(\frac{1}{s(s+u)}\right)^{\frac{k}{2}}\mathcal{M}_{\text{low}}(s,u)\right]_{s=0}$$

$$=\left\langle\left(\frac{2m^2+u}{(m^2+u)}\right)\left(\frac{1}{m^2(m^2+u)}\right)^{\frac{k}{2}}\tilde{C}_J^{\left(\frac{d-3}{2}\right)}\left(1+\frac{2u}{m^2}\right)\right\rangle,$$

$$\langle f(m^2,J)\rangle \equiv \frac{1}{2\pi}\sum_{J\text{ even}}\mathcal{N}_J\int_{M^2}^{\infty}dm^2\frac{m^{4-d}}{m^2}2\text{Im}[c_{m^2,J}]\left[f(m^2,J)\right].$$

- $\langle - \rangle$ defines a linear functional with positive measure: $0 \leq \text{Im}\left[c_{s,J}\right]$
- EFT data = averages in the UV

Bootstrap/B_k sum rules

Taylor expanding the Gegenbauers produces self-consistency conditions

$$B_2: \quad 2g_2 - g_3 u + 8g_4 u^2 + \ldots \quad = \left\langle \frac{\left(2m^2+u\right)\tilde{C}_j\left(1+\frac{2u}{m^2}\right)}{m^2\left(m^2+u\right)^2} \right\rangle$$

$$B_4: \quad 4g_4 + \ldots \quad = \left\langle \frac{\left(2m^2+u\right)\tilde{C}_j\left(1+\frac{2u}{m^2}\right)}{m^4\left(m^2+u\right)^3} \right\rangle.$$

and matching order-by-order when Taylor expanding in u generates a list of identities!

Bootstrap/B_k sum rules

Identities on the couplings

$$g_2 = \left\langle \frac{1}{m^4} \right\rangle > 0$$

$$g_3 = \left\langle \frac{3 - \frac{4}{d-2}\mathcal{J}^2}{m^6} \right\rangle$$

$$g_4 = \left\langle \frac{4 + \frac{2(4-5d)}{d(d-2)}\mathcal{J}^2 + \frac{4}{d(d-2)}\mathcal{J}^4}{m^8} \right\rangle$$

$$g_4 = \left\langle \frac{1}{2m^8} \right\rangle > 0$$

and produces a **null constraint** by writing g_4 in two different ways

$$0 = \left\langle \frac{\mathcal{J}^2(4 - 5d + 2\mathcal{J}^2)}{m^8} \right\rangle$$

Bootstrap/Physical interpretation of B_k sum rules

- EFT data $g_n \leftrightarrow \langle$particle production in the UV$\rangle_{m,J}$

$$g_3 = \left\langle \frac{3 - \frac{4}{d-2}\mathcal{J}^2}{m^6} \right\rangle$$

- Positive linear functional $\implies \langle$positive quantity$\rangle > 0$

$$g_4 = \left\langle \frac{1}{2m^4} \right\rangle > 0$$

- Null constraints tell us it is possible to relate the $\mathcal{J}^4$ averages with averages independent of $\mathcal{J}$

$$0 = \left\langle \frac{\mathcal{J}^2(4 - 5d + 2\mathcal{J}^2)}{m^8} \right\rangle$$

- Produces an ∞ collection of $B_2, B_4, \cdots$ and identities on g_n's
- Possible to add ∞ more **null constraints** using this scheme

Bootstrap/Strategy

Outline of the bootstrap

1. Write the polynomial expansion for $\mathcal{M}_{\text{low}}(s,t,u)$ to high order
2. Relate the IR and UV expansion using dispersion relations to derive the B_k sum rules
3. Solve an optimization problem
4. Infer bounds on the couplings

$$\frac{\#}{M^2} \leq \frac{g_3}{g_2} \leq \frac{\#}{M^2}, \qquad \frac{\#}{M^4} \leq \frac{g_4}{g_2} \leq \frac{\#}{M^4}, \quad \ldots$$

So far we have performed steps 1-2. We now present the optimization problem

Optimization problem/Definition

Goal: Find the **optimal** lower bound $A \leq \tilde{g}_3 = g_3 \frac{M^2}{g_2}$ using

- identities on g_n's in terms of $\langle - \rangle$ in the UV
- null constraints

Solve

$$\begin{aligned} \text{maximize:} \quad & A \\ \text{subject to:} \quad & 0 \leq -Ag_2(m^2, J) + M^2 g_3(m^2, J) + \text{null constraints} \end{aligned}$$

Finding such a number A and a particular linear combination of null constraints

$$0 \leq -Ag_2 + M^2 g_3 + 0 \implies A \leq \tilde{g}_3 \equiv g_3 \frac{M^2}{g_2}$$

So what? We can enhance the bound by including more functions in the null constraints

Optimization problem/In practice

1. Increase the order of $\mathcal{M}_{\text{low}}$

$$\mathcal{M}_{\text{low}} = \cdots + g_n(s^2 + t^2 + u^2)^a (stu)^b$$

2. Increase the number of accessible B_k sum-rules

$$\text{Res}\left[\#\mathcal{M}_{\text{low}}\right] = \langle \#\text{Gegenbauer} \rangle$$

3. Increase the number of null constraints

$$0 = \langle \text{complicated function in UV} \rangle_{m,J}$$

4. Improve the bound on the couplings g_n's

$$\begin{aligned} \text{maximize:} \quad & A \\ \text{subject to:} \quad & 0 \leq -A g_2(m^2, J) + M^2 g_3(m^2, J) + \text{null constraints} \end{aligned}$$

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Numerically ruling-out/Results in 3+1 spacetime

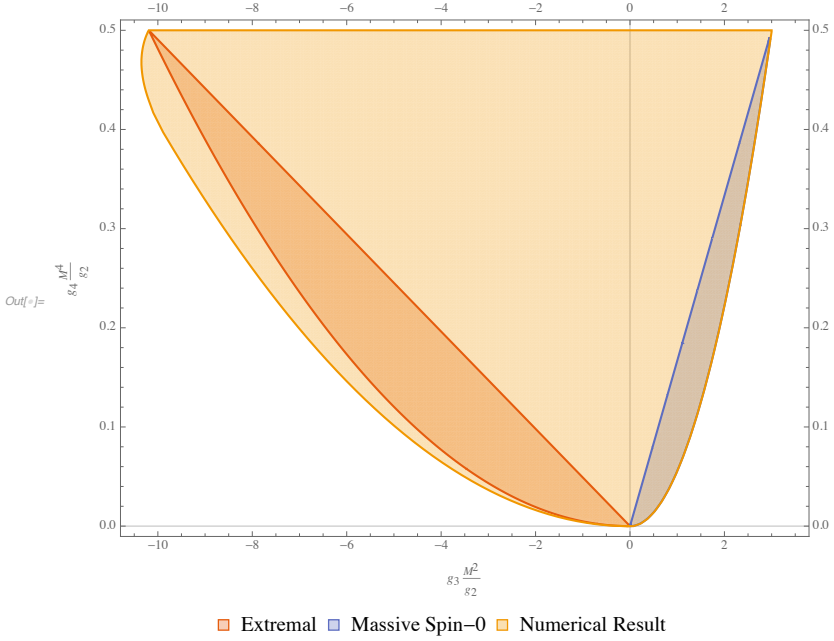
Results from numerics:

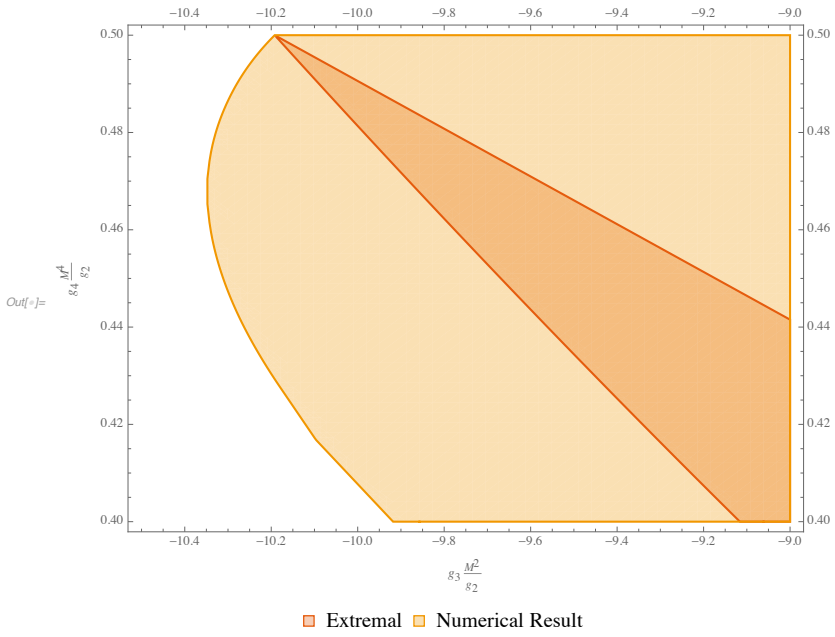
- Basic bounds on couplings

$$\frac{-10.35}{M^2} \leq \frac{g_3}{g_2} \leq \frac{3}{M^2}$$
$$0 \leq \frac{g_4}{g_3} \leq \frac{\frac{1}{2}}{M^4}$$
$$-\frac{-4.09}{M^6} \leq \frac{g_5}{g_2} \leq \frac{\frac{5}{2}}{M^6}$$

- Not all vectors of g_n's are allowed (i.e. the space of coefficients forms a non-trivial convex hull)
- There is a simple explanation for this convex hull shape

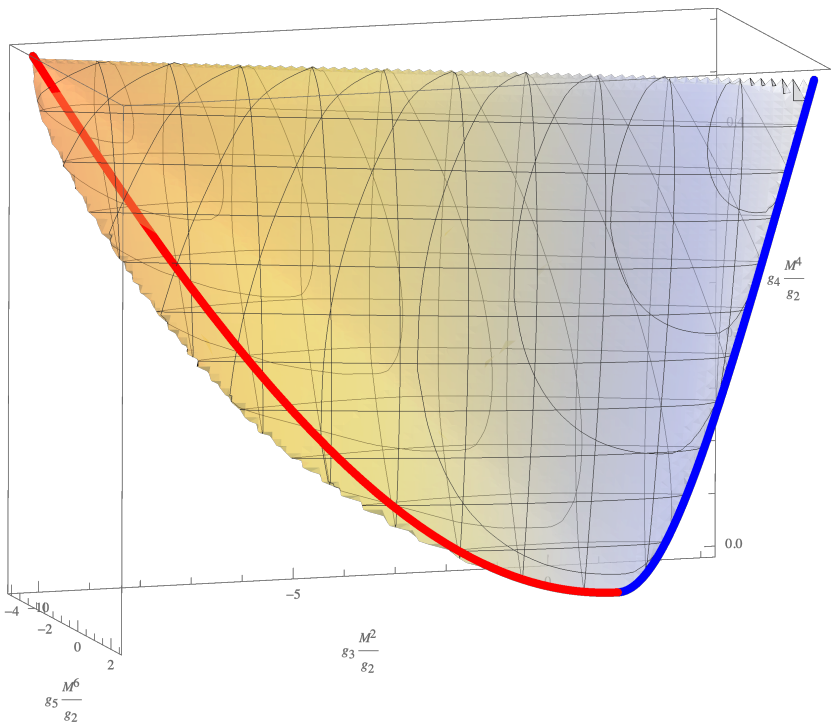
Numerically ruling-out/Exclusion Plots

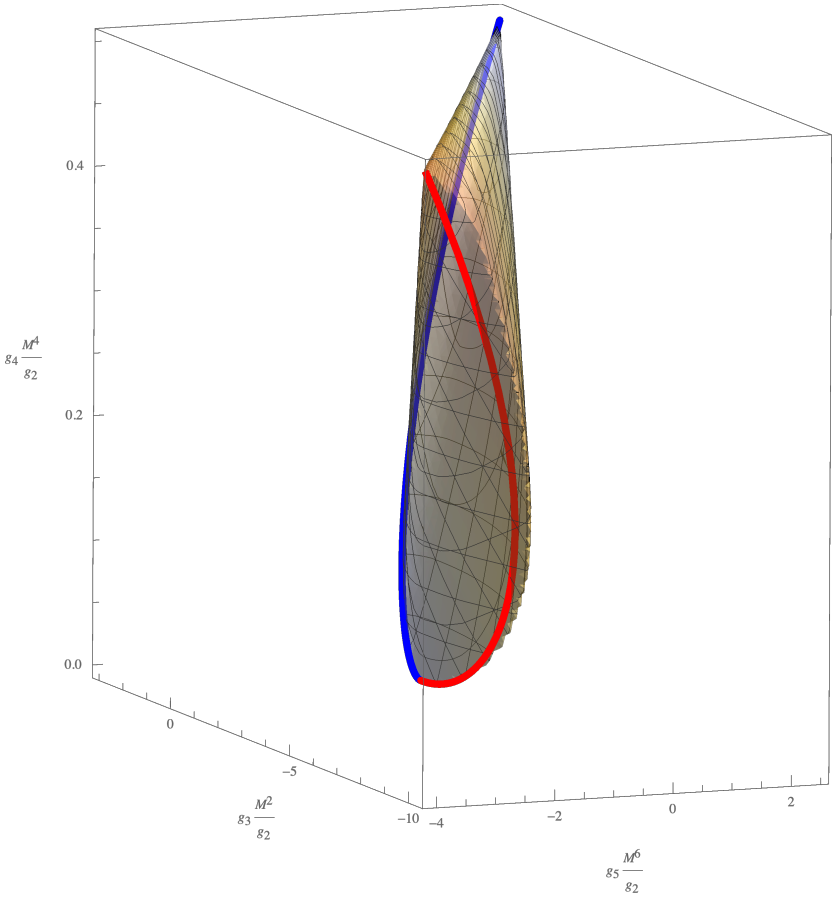




- Ensuring positivity yields on the spin-0 particles $\rightarrow$ **right** underbelly
- Ensuring positivity on spin-$2, 4, 6, \cdots \rightarrow$ **left** underbelly
- Taking the convex hull generates the entire set
- Kinks: $(\tilde{g}_3, \tilde{g}_4) = (3, 0.5), \qquad (\tilde{g}_3, \tilde{g}_4) = (-10.19, 0.5)$

Numerically ruling-out/Exclusion Plots





- We can keep carving out the space for higher order g_n's, but it's computationally expensive at the moment
- In theory, the allowed EFT coefficients g_n will form an infinite dimensional convex polytope with infinitely-many faces

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Analytically ruling-in/Reverse-engineering the kinks

The following amplitudes reproduce the kinks when $m = M$

$$\mathcal{M}_{\text{spin-0}} = \left[\frac{1}{m^2 - s} + \frac{1}{m^2 - t} + \frac{1}{m^2 - u}\right]$$

$$\mathcal{M}_{stu\text{-pole}} = \frac{m^4}{(s - m^2)(t - m^2)(u - m^2)} + \gamma(d)\mathcal{M}_{\text{spin-0}}$$

where $\gamma(d) \equiv \frac{4}{9}\, {}_2F_1\left(\frac{1}{2}, 1, \frac{d-1}{2}, \frac{1}{9}\right)$

- spin-0:

$$(\tilde{g}_3, \tilde{g}_4, \tilde{g}_5) = \left(\frac{3M^2}{m^2}, \frac{M^4}{2m^4}, \frac{5M^6}{2m^6}\right)$$

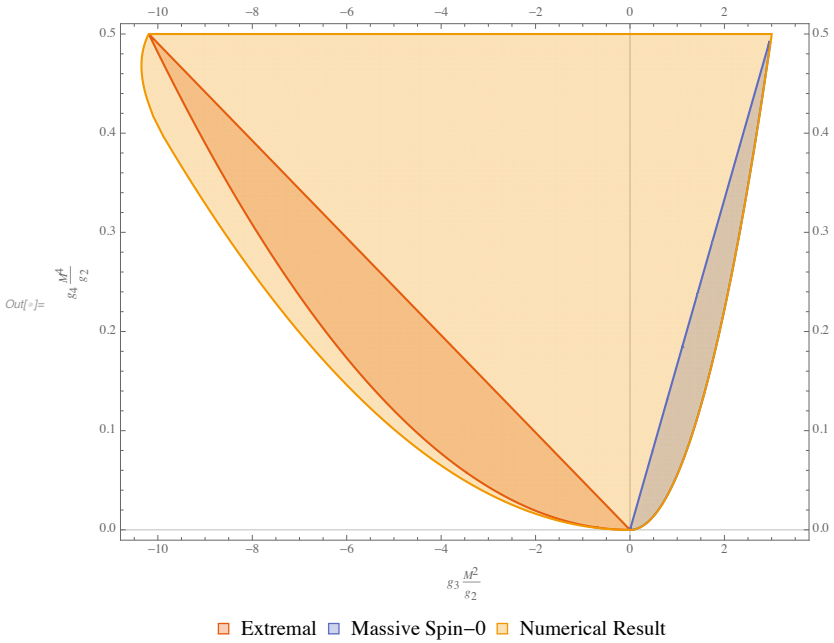
- stu-pole:

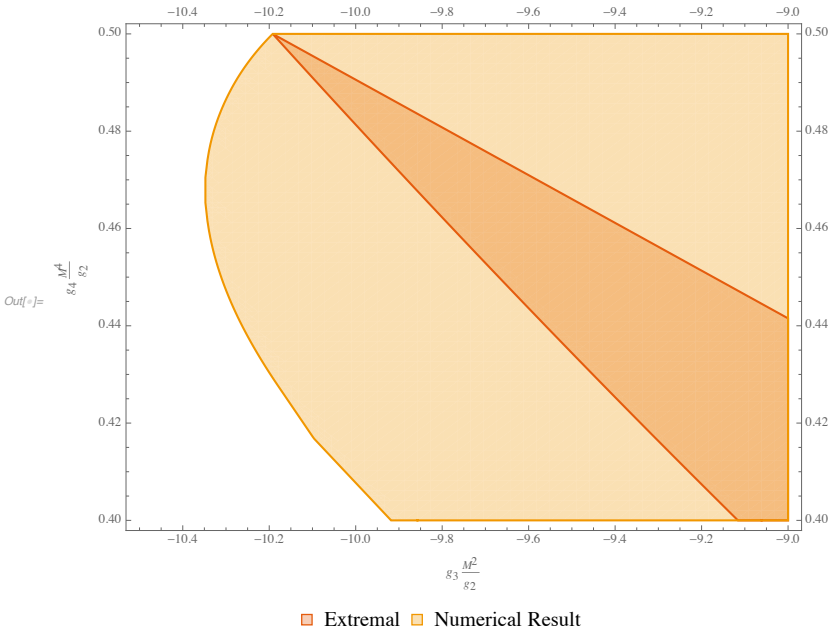
$$(\tilde{g}_3, \tilde{g}_4, \tilde{g}_5) = \left(\frac{2(3\gamma(d) - 1)M^2}{(2\gamma(d) - 1)m^2}, \frac{M^4}{2m^4}, \frac{(5\gamma(d) - 2)M^6}{(2\gamma(d) - 1)m^6}\right)$$

Analytically ruling-in/Reverse-engineering the kinks

Re-scaling the mass gap $m/M > 1$ traces out curves in the exclusion plot that corresponds to the convex hull generated

$$\text{Allowed couplings} = \text{Convex hull of}\,[\mathcal{M}_{\text{spin-0}} + \mathcal{M}_{stu\text{-pole}} + \text{Remaining slice}]$$





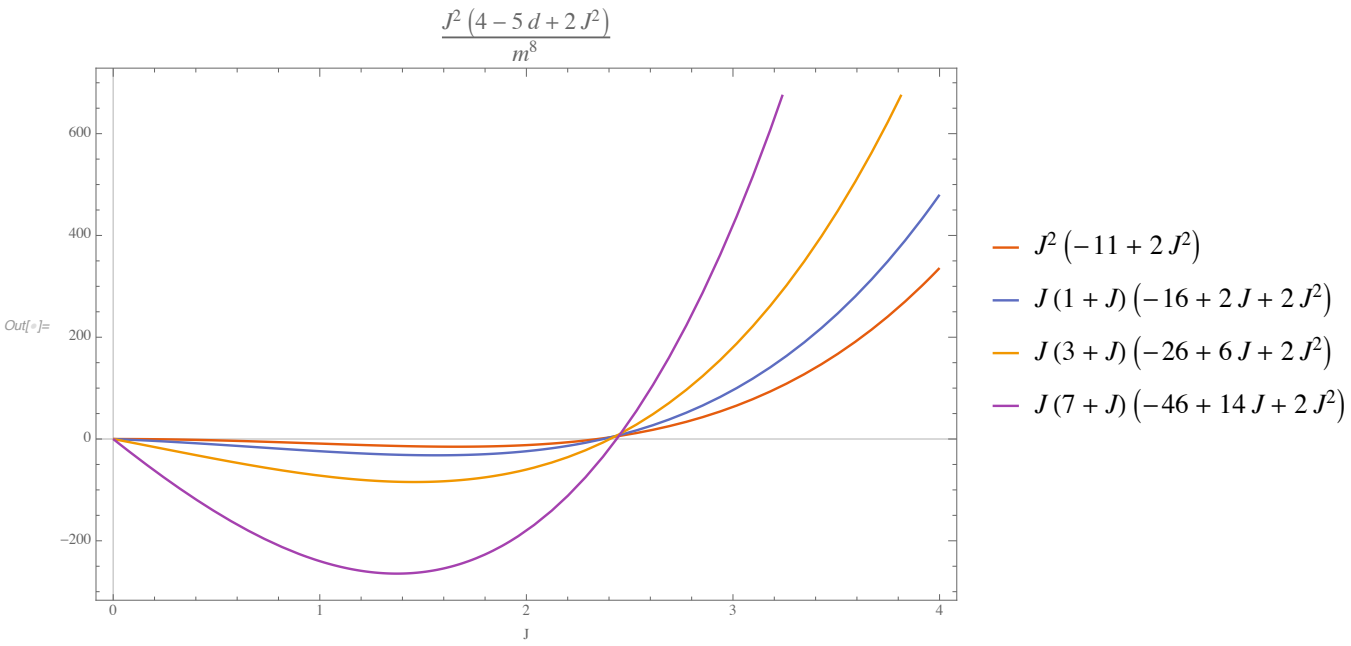
Analytically ruling-in/Simple results from crossing

With only a single null constraint arising from g_4, we can make some statements on the EFT

$$
\begin{aligned}
0 &= \left\langle \frac{\mathcal{J}^2(4-5d+2\mathcal{J}^2)}{m^8} \right\rangle \\
&= \left\langle \frac{J(J+d-3)\,(4-5d+2J(J+d-3))}{m^8} \right\rangle
\end{aligned}
$$

- Physical roots located between $2 \leq J \leq 3$
- In order to average to 0, the function requires to change signs at different values of J
- That means these UV processes necessarily have **spin-2** particles + **spin-4** or higher
- **So what?** If the theory contains spinning particles, then it contains a $J = 2$ particle

Analytically ruling-in/Simple results from crossing



- $0 = \left\langle \frac{\mathcal{J}^2(4-5d+2\mathcal{J}^2)}{m^8} \right\rangle$
- In any space-time dimensions, the physical roots are located between $2 \leq J \leq 3$
- **So what?** If the theory contains spinning particles, then it contains a $J = 2$ particle + $J \geq 4$ particle(s)

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Summary of results

1. Dispersion relations on $\mathcal{M}$ connect the IR and UV behaviour
2. Positivity defines an averaging technology to build inequalities:

$$0 \leq \text{Im}\left[\mathcal{M}\right] \implies 0 \leq \langle \text{positive quantity} \rangle$$

3. EFT data $g_n \leftrightarrow \langle \text{particle production in UV} \rangle_{m,J}$
4. Null constraints are non-trivial and rich in data

$$0 = \left\langle \frac{\mathcal{J}^2(4-5d+2\mathcal{J}^2)}{m^8} \right\rangle = \cdots = \langle \text{complicated functions} \rangle_{m,J}$$

5. Numerically optimizing allows us to carve-out the EFT
6. EFT exclusion region = Convex hull of (spin-0 theory + spinning theories)
7. Kinks in numerics may be reverse engineered using analytical amplitudes: $\mathcal{M}_{\text{spin-0}} + \mathcal{M}_{stu\text{-pole}}$

Next questions

There are still important questions to answer:

1. Is it possible to use the full positivity condition to derive stronger bounds?

$$0 \leq \mathsf{Im}\left[c_{s,J}\right] \leq 1$$

   - We only used the lower bound
   - Is it possible to use the upper bound too?

2. Is there an analytical amplitude like $\mathcal{M}_{\text{spin-0}}$ & $\mathcal{M}_{stu\text{-pole}}$ that will complete the region in the numerical exclusion plot?

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2D Exclusion Region

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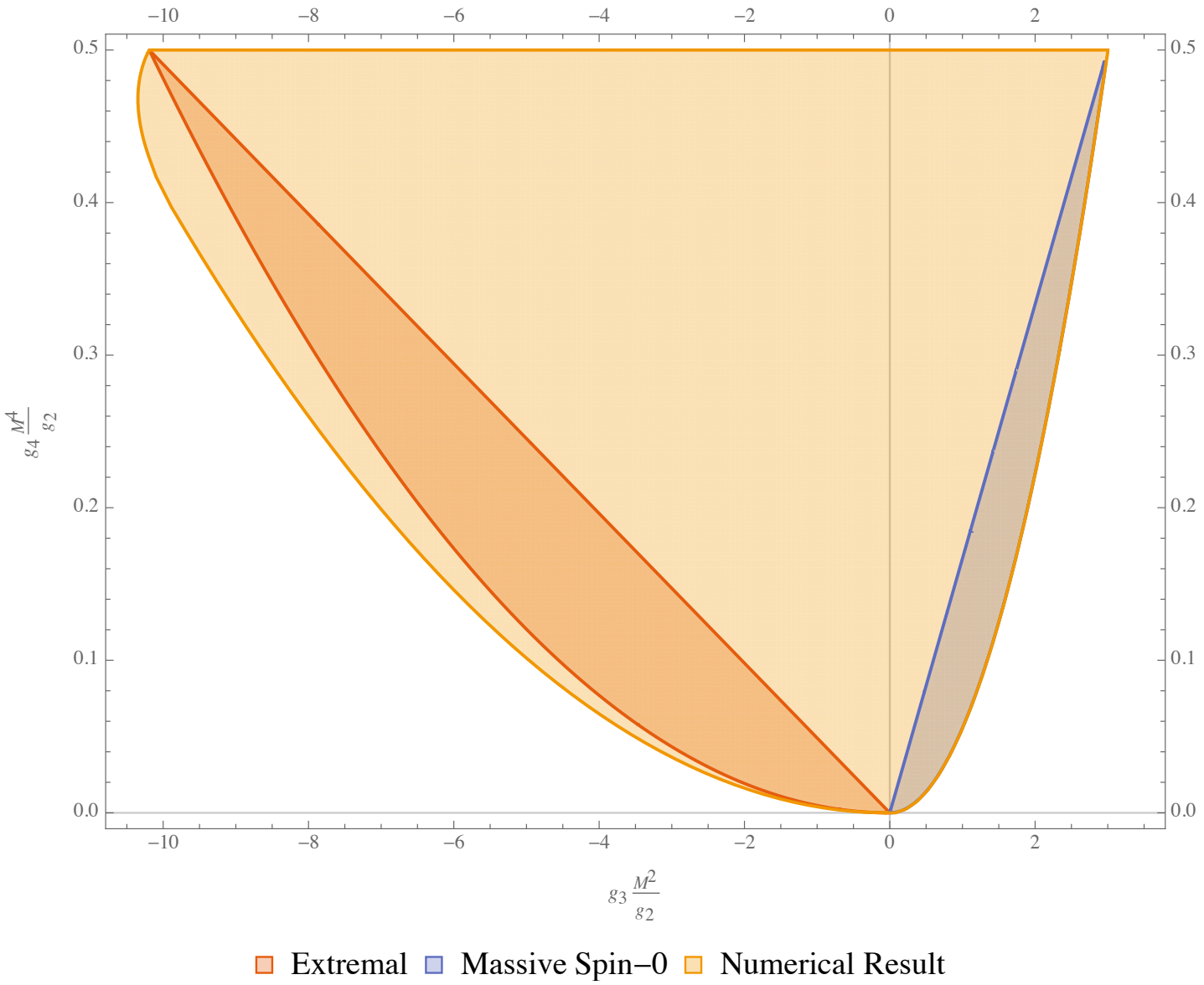

2D Exclusion Region (zoom)

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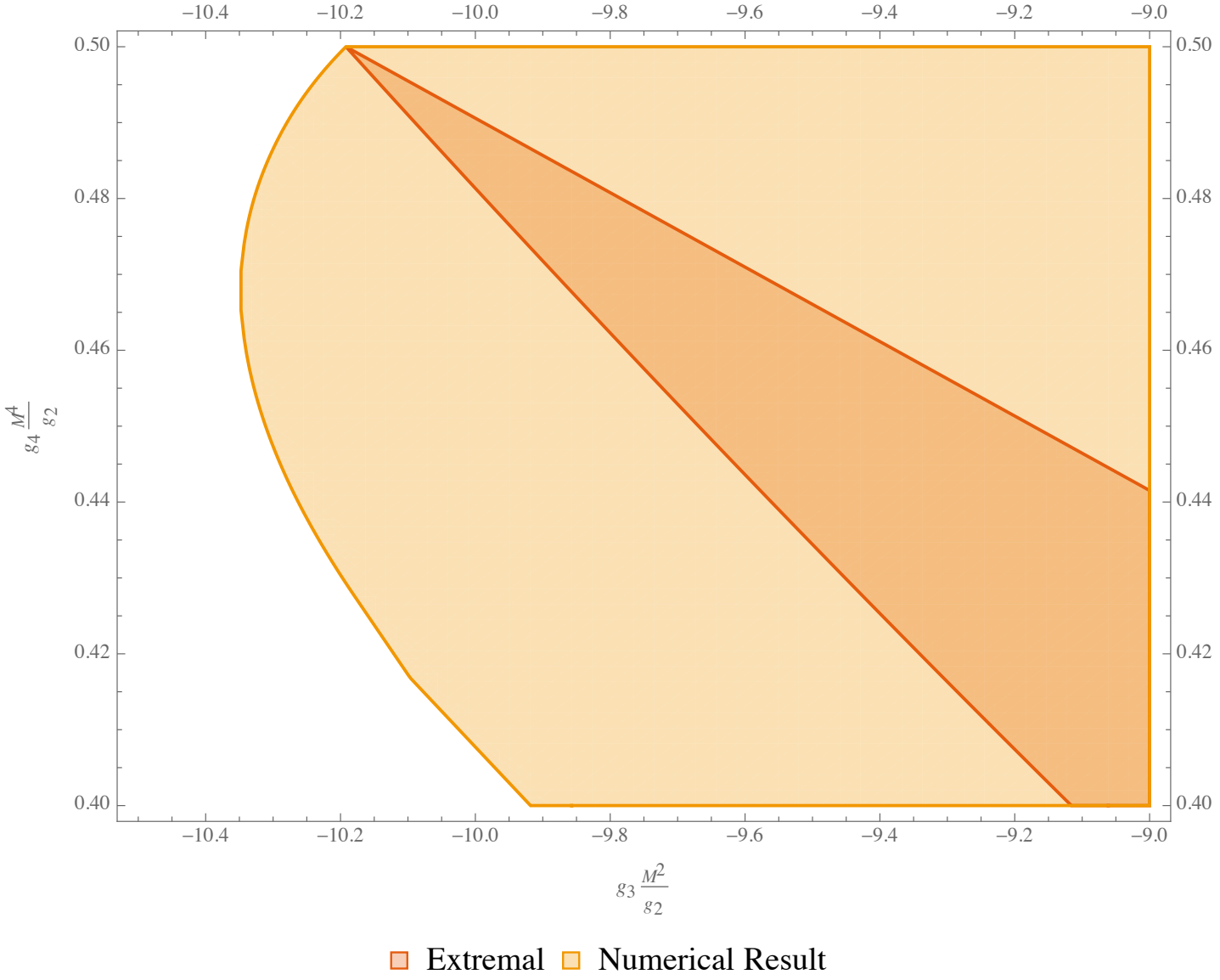
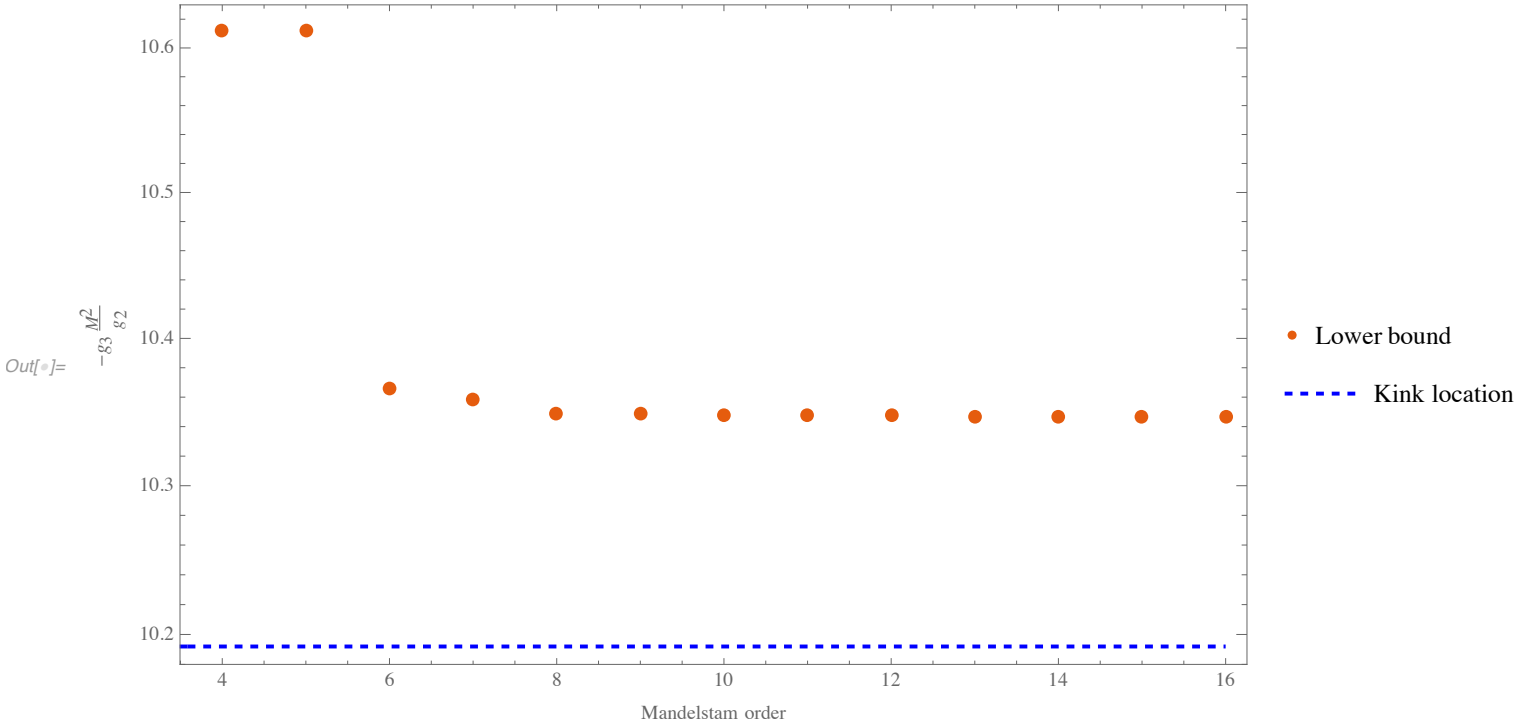
Convergence